

- 7 (a) α -particle: *either* helium nucleus *or* contains 2 protons + 2 neutrons
or ${}^4_2\text{He}$ B1
- β -particle: *either* electron *or* ${}^0_{-1}\text{e}$ B1
- α speed < β speed (1)
- α discrete values of speed/energy, β continuous spectrum (1)
- either* α ionising power $\gg \beta$ ionising power
- or* α range $\ll \beta$ range (1)
- α positive, β negative (*only if first two B marks not scored*) (1)
- α mass > β mass (*only if first two B marks not scored*) (1)
- (*any two sensible pairs of statements relevant to differences,*
– do not allow statements relevant to only α or β , 1 each, max 2) B2 [4]
- (b) (i) ${}^{236}_{92}\text{U} \rightarrow {}^{232}_{90}\text{Th}$ M1
 $+ {}^4_2\text{He}$ A1 [2]
- (ii) 1. correct position for U at $Z = 92$, $N = 145$ B1
 2. correct position for Np relative to U i.e. $Z + 1$ and $N - 1$ B1 [2]